
Products and Quotients of Rational Expressions

Multiply

**Multiply an Integer or a
Polynomial**

Divide

$$\frac{3x}{x+1} \cdot \frac{x^2+x}{3x-6}$$

$$\frac{x+2}{x^2-4} \cdot (x^2-2x) = \frac{x+2}{x^2-4} \cdot \frac{x^2-2x}{1}$$

$$\frac{1-x^2}{x^2+3x-4} \div \frac{x+1}{x+4} =$$
$$\frac{1-x^2}{x^2+3x-4} \cdot \frac{x+4}{x+1}$$

The domain is $x \neq -1$ or 2 .

The domain is $x \neq -2$ or 2 .

The domain is $x \neq -4, -1$, or 1 .

Identify common factors and
simplify.

Write the polynomial as a rational
expression with 1 in the denominator.
Then multiply.

Multiply by the reciprocal of the
divisor.
